

Reviewer Pack — Minimal Order of Monoid Actions and Communication Complexity...

Entry 43b9978834e9 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Order of Monoid Actions and Communication Complexity Rank Inequality

Entry ID: 43b9978834e9 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-05 05:17:38 UTC

1. Conjecture

Field A (mathematical branch): Algebra (Monoids) **Field B** (complexity object): Communication Complexity (Matrix Rank)

Statement:

For every instance φ of size n , the minimal order of a monoid action on the symmetric group S_n associated with φ is linearly correlated with its communication complexity rank $r(\varphi)$, such that $O(\log n) \leq m(\varphi) \leq n!$

Rationale (proposer's reasoning):

Monoids have been studied in algebraic combinatorics and their actions can provide structural insights into combinatorial problems. By considering the minimal order of a monoid action, we might uncover hidden structures related to communication complexity that could lead to new understanding or lower bounds.

Taxonomy category: MonoidActions (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 68a84e4536f7e501

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if at least 80% of instances φ show a correlation coefficient between the minimal order $m(\varphi)$ and communication complexity rank $r(\varphi)$ within the range $[0.8, 1]$. The conjecture is falsified if any instance violates this correlation by having a correlation coefficient outside this range.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "monoid actions" AND "communication complexity rank" AND "symmetric group S_n " - "minimal order of monoids" related to "matrix rank in communication complexity" - "linear correlation" between "monoid action on S_n " and "communication complexity rank"

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=0, elapsed=0.0s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def factorial(n):
        if n == 0 or n == 1:
            return 1
        result = 1
        for i in range(2, n + 1):
            result *= i
        return result

    def communication_complexity_rank(n):
        # Placeholder function to compute the communication complexity rank
        # This is a dummy implementation; replace with actual computation if available
        return random.randint(1, n)

    def minimal_order_of_monoid_action(n):
        # Placeholder function to compute the minimal order of monoid action
        # This is a dummy implementation; replace with actual computation if available
        return random.randint(1, factorial(n))

    n = random.choice([5, 10, 15, 20, 30, 40])
    m_phi = minimal_order_of_monoid_action(n)
    r_phi = communication_complexity_rank(n)

    correlation_coefficient = (m_phi - r_phi) / math.sqrt(m_phi**2 + r_phi**2)

    return {
        "metric_name": "correlation_coefficient",
        "metric_value": correlation_coefficient,
```

```

        "instances_tested": 1,
        "n_max": n,
        "conjecture_holds": abs(correlation_coefficient) >= 0.8 and abs(correlation_coefficient) <
        "counterexample": ""
    }

if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3
    results = []

    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {{\"seed\": {seed}, \"metric_name\": \"{result['metric_name']}\"}, \"metric_value\": {result['metric_value']}}")
        results.append(result)

    mean_metric_value = sum(r['metric_value'] for r in results) / len(results)
    std_metric_value = math.sqrt(sum((r['metric_value'] - mean_metric_value) ** 2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r['conjecture_holds']) / len(results)

    if all(r['conjecture_holds'] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_metric_value} support_fraction={support_fraction}")
    elif any(not r['conjecture_holds'] for r in results) and support_fraction >= 0.8:
        first_failing_seed = next(s for s, r in zip(seeds, results) if not r['conjecture_holds'])
        print(f"RESULT: FALSIFIED counterexample=\"correlation_coefficient_outside_range\" first_failing_seed={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE insufficient_data")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

0.9999975619359691, "instances_tested": 1, "n_max": 10, "conjecture_holds": True, "counterexample": ""
TRIAL: {"seed": 503, "metric_name": "correlation_coefficient", "metric_value": 0.9999905733744969, "instances_tested": 1}
TRIAL: {"seed": 547, "metric_name": "correlation_coefficient", "metric_value": 0.6704783996548059, "instances_tested": 1}
TRIAL: {"seed": 593, "metric_name": "correlation_coefficient", "metric_value": 0.9999999999854784, "instances_tested": 1}
TRIAL: {"seed": 631, "metric_name": "correlation_coefficient", "metric_value": 0.999999999973605, "instances_tested": 1}
TRIAL: {"seed": 677, "metric_name": "correlation_coefficient", "metric_value": 1.0, "instances_tested": 1}
TRIAL: {"seed": 727, "metric_name": "correlation_coefficient", "metric_value": 1.0, "instances_tested": 1}
TRIAL: {"seed": 773, "metric_name": "correlation_coefficient", "metric_value": 0.9999971165546288, "instances_tested": 1}
TRIAL: {"seed": 821, "metric_name": "correlation_coefficient", "metric_value": 1.0000000000000002, "instances_tested": 1}
TRIAL: {"seed": 877, "metric_name": "correlation_coefficient", "metric_value": 0.9999978872870667, "instances_tested": 1}
TRIAL: {"seed": 929, "metric_name": "correlation_coefficient", "metric_value": 0.9999991015919791, "instances_tested": 1}
RESULT: FALSIFIED counterexample="correlation_coefficient_outside_range" first_failing_seed=53

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The test code uses dummy implementations for both the minimal order of a monoid action and the communication complexity rank, which are crucial components of the conjecture. These placeholders do not provide meaningful empirical evidence to support or refute the conjecture.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: The test code uses dummy implementations for both the minimal order of a monoid action and the communication complexity rank, which are crucial compon | next: Replace dummy implementations with actual calculations for both the minimal order of a monoid action and the communication complexity rank to conduct a meaningful test.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	12030
2	preregistration	ollama_remote	glm4:latest	0	0	9433
3	novelty	ollama_remote	glm4:latest	0	0	8509
4	novelty	ollama_remote	glm4:latest	0	0	8957
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13951
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10358
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10952
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8499
9	critic	ollama_remote	glm4:latest	0	0	13917
10	judge	ollama_remote	glm4:latest	0	0	9687

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 106293 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in research/audit/43b9978834e9.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/43b9978834e9.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/43b9978834e9.tar.gz (if generated)